

Total No. of Questions : 8]

SEAT No. :

PB-2227

[Total No. of Pages : 2

[6263]-64

B.E. (Civil)

AIRPORT AND BRIDGE ENGINEERING
(2019 Pattern) (Semester - VII) (401004D) (Elective - IV)

Time : 2½ Hour]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6 and Q7 or Q8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) Explain the necessity of airport drainage. [6]
b) What is BIM? Explain its benefits. [6]
c) Give comparison between the Highway and Airfield pavement. [6]

OR

- Q2)** a) Explain the factors affecting the design of Airport capacity. [6]
b) Briefly explain the application of Building Information Modelling (BIM) in airport engineering. [6]
c) Write a note on planning and design of airport pavement. [6]

- Q3)** a) What is the need of runway lightening and marking? [6]
b) Explain with neat sketch - [6]
i) Approach Lights
ii) Runway End Lights
iii) Taxiway Lights
c) Explain the type of runway marking. [6]

OR

- Q4)** a) Draw the neat sketch of Heliport orientation. [6]
b) Explain vertical take-off and landing, short take-off and Landing. [6]
c) What is Runway side strips marking? [6]

P.T.O.

- Q5)** a) Give detailed classification of the bridges. [6]
b) What are the characteristics of ideal bridge site? [6]
c) What is Afflux? Give Maririman's and Molesworth's equation to calculate afflux. [5]

OR

- Q6)** a) Enlist the design loads to be considered in case of Bridge. [6]
b) Explain the concept of linear waterway. [6]
c) Write short note on economic span of bridge. [5]

- Q7)** a) What is Culvert? Give its types. [6]
b) Explain in details Temporary bridges. [6]
c) Compare the salient features of Simply supported, continuous and cantilever bridges. [5]

OR

- Q8)** a) Give the functions of bearing in Bridges. Explain any one type of bearing with figure. [6]
b) What is moveable bridge? Enlist all types. Explain Any one [6]
c) Give factors affecting the location of the Culvert. [5]

